Predicting Song Popularity

How record labels can use audio features and listener data to evaluate track potential.

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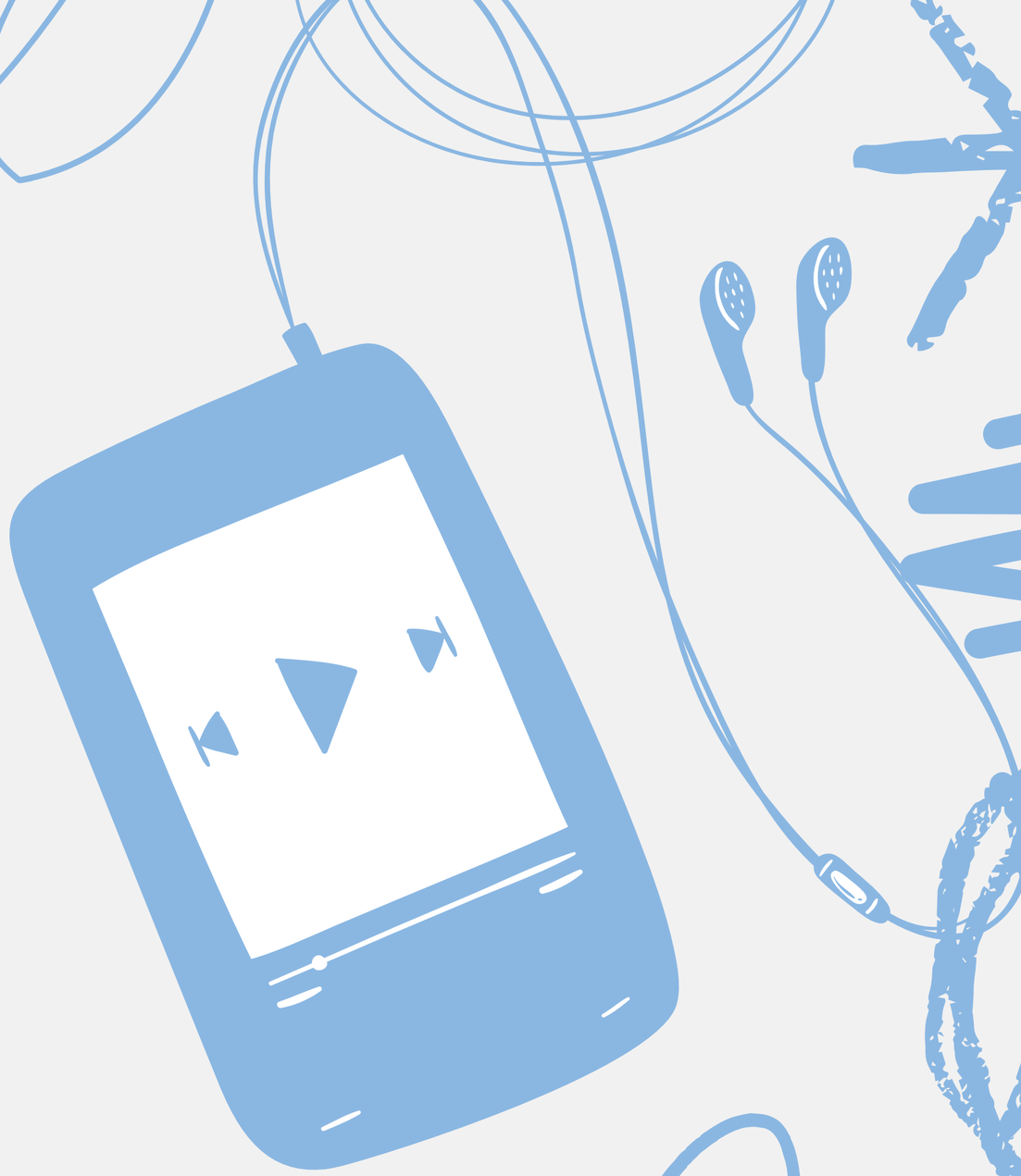


Intro

Analysis Overview

TARGET AUDIENCE

A&R and marketing teams at record labels looking for a more data-informed way to evaluate track potential.

BUSINESS QUESTION

Can record labels use song traits to better predict which tracks are likely to become popular?

WHY THIS MATTERS

Explore whether audio features and listener engagement provide useful signals for predicting track popularity

DATA

Last.fm API provided listener engagement data; Spotify audio features (Kaggle) provided track characteristics and popularity measures.

TESTING

Compared song audio traits with listener engagement and Spotify popularity to see whether certain track profiles appear more often among stronger-performing songs.

KEEP IN MIND

Because the Last.fm–Spotify match was string-based, only about 51.7% of Last.fm tracks matched to Spotify features, so results should be used as directional guidance rather than exact prediction.

Python Data Analysis

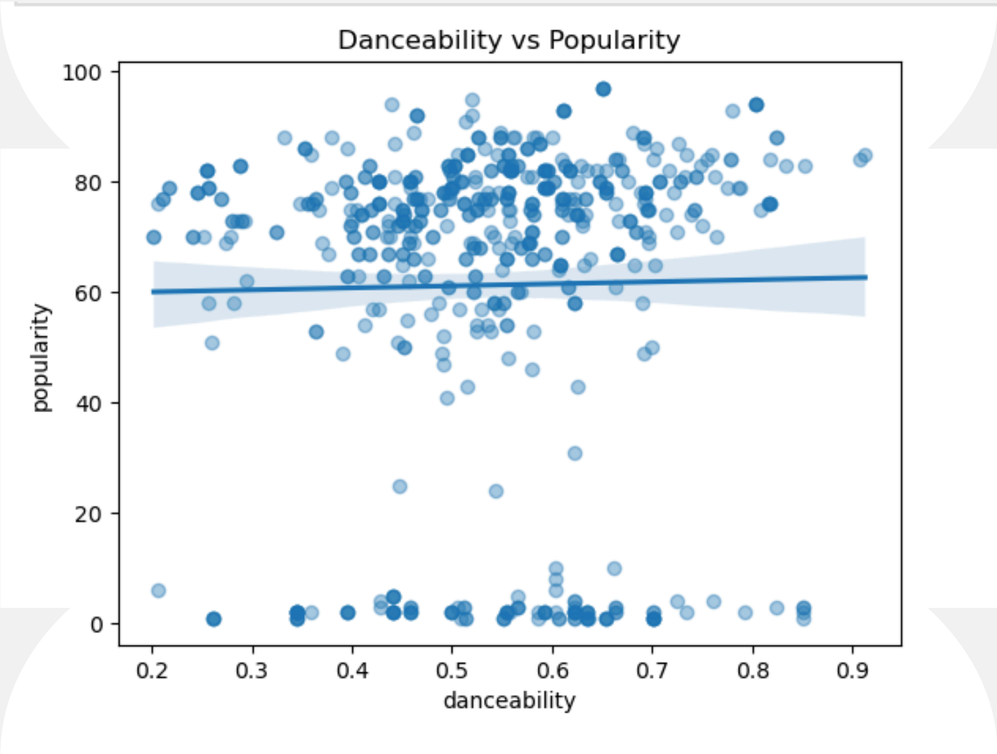
Clean + Merge/ Data Validation

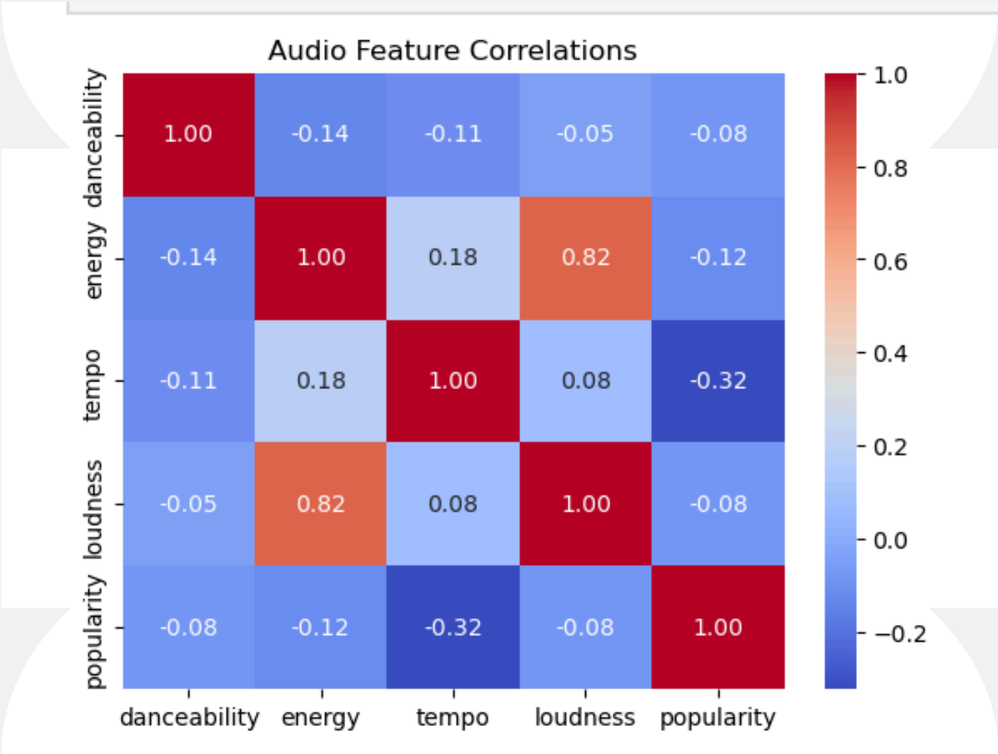
Standardized artist and track names, used a left join to merge Last.fm with Spotify, and checked duplicates, missing Spotify fields, match coverage, and summary statistics.

Feature Exploration

Built a popularity distribution, tested danceability vs. popularity, created a correlation heatmap for major audio variables, and compared artist- and tempo-level patterns across matched songs.

Key Data Patterns

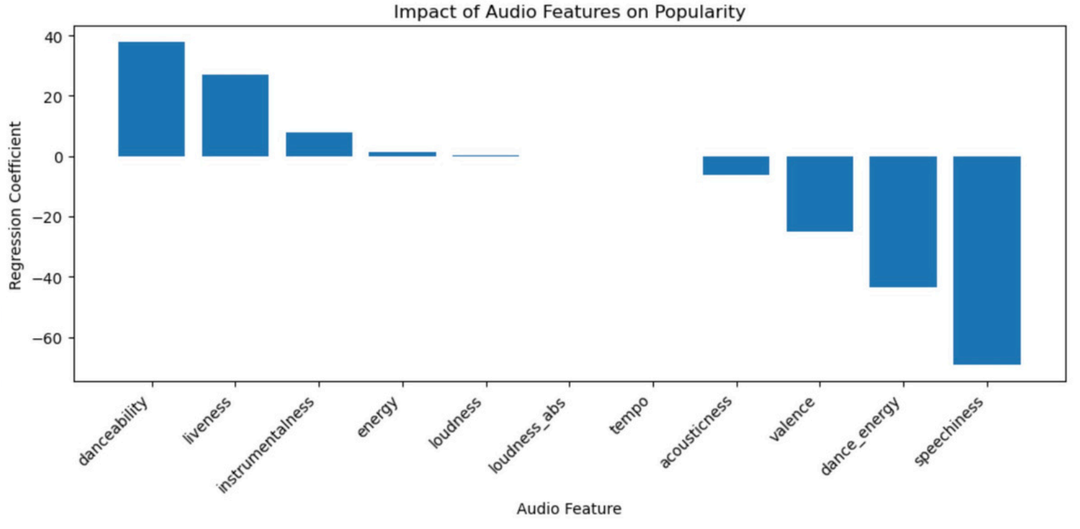




Danceability shows a slight positive relationship with popularity, but the trend is weak.

Most audio features have weak correlations with popularity, suggesting that song success depends on more than sound alone.

Regressions



Linear Regression

Goal: Predict song popularity score

- **Dependent variable:** Spotify popularity
- **Predictors:** danceability, energy, tempo, loudness

Model Performance

- **R^2:** 0.1299
- **RMSE:** 30.9

Key finding

- The model explained only a small share of variation in popularity
- Overall fit remained low, so individual coefficients should be interpreted cautiously

<u>What this means for labels:</u>

Audio features may provide some directional insight, but they are not strong enough to evaluate track potential on their own.

Regressions

Logistic Regression

Goal: Classify songs as high vs low popularity

- **Predictors:** danceability, energy, tempo, loudness

Model Performance

- **Accuracy:** 0.62
- **ROC-AUC:** 0.62
- **F1 Score:** 0.695

Result

- The model correctly classified songs about **62%** of the time
- Performance was modest overall, suggesting only limited separation between higher- and lower-popularity songs

<u>What this means for labels:</u>

Audio features may help with early screening, but they are not strong enough to be used as a standalone decision tool.

SQL Data Analysis

Clean + Merge/ Data Validation

Loaded Spotify and Last.fm tables separately into SQLite, built cleaned join keys, and used INNER JOINs to identify matched songs and a LEFT JOIN to evaluate artist-level match coverage.

Feature Exploration

Used grouped queries and window functions to rank artists and songs, compare listener counts, playcounts, and average popularity, and identify which matched tracks showed the strongest cross-platform traction for label attention.

What This Means for Record Labels

- **Audio features provide some directional signal:** The analysis suggests that certain audio traits may be modestly associated with popularity.
- **Audio traits alone are not enough:** Popularity is also shaped by marketing, playlist placement, timing, and artist visibility.
- **Best use case for labels:** Use audio and listener data as an early screening tool, not as a standalone predictor of success.
- **Recommendation:** Combine song features with broader market and artist data when deciding which tracks to prioritize.

Thanks!